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ORIGINAL ARTICLE

Evidence-based recommendations for the diagnosis, treatment, functional rehabilitation, and follow-up of people over 7 years of age with low vision

Recomendaciones basadas en la evidencia para el diagnóstico, el tratamiento, la rehabilitación y el seguimiento de personas mayores de 7 años con baja visión

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Abstract

Purpose: *The aim of the study was to develop evidence-based recommendations for the diagnosis, functional rehabilitation, and follow-up of people with low vision between 7 and 18 years of age and > 18 years of age.* **Methods:** *A development group was formed with the participation of patients. A systematic search was performed, and the quality of the studies and the body of evidence were evaluated. Recommendations were made using the Grade Evidence to Decision Framework, and then validated by national experts.* **Results:** *In the diagnostic section, the quality of evidence regarding visual acuity and visual field was low; in these tests, a weak recommendation was made favorable to the use of the Early Treatment Diabetic Retinopathy Study and Microperimetry primer, respectively. Regarding contrast sensitivity, the quality of evidence was very low, and a weak recommendation was given favorable to the use of the Pelli-Robson test. Regarding treatment, the quality of evidence on electronic optical aids was moderate, and a weak recommendation was given favorable to their use. Regarding manual and electronic magnifiers, telescopes, and tablets, the quality of evidence was very low and a weak recommendation favorable to their use was given. In the rehabilitation section, the quality of evidence regarding this aspect for people > 16 years of age was moderate, and there is a strong recommendation favorable to multidisciplinary and multicomponent rehabilitation. On the other hand, the quality of evidence regarding rehabilitation for people between 7 and 16 years of age was low, and there is a strong recommendation favorable to its use. The quality of evidence for occupational therapy interventions was low and a weak recommendation was given favorable of its use. Regarding psychology, the quality of evidence on interventions like this was low and a strong recommendation was given favorable to their use. Finally, the quality of evidence for the use of follow-up scales was high and a strong recommendation was given favorable to the use of the National Eye Institute Visual Function Questionnaire-25.* **Conclusions:** *The recommendations issued facilitate establishing general practice guidelines for the comprehensive management of low vision to reduce unwanted variability regarding diagnosis, functional rehabilitation, and follow-up.*

Keywords: *Low vision. Clinical practice guideline. Rehabilitation. Assistive technologies. Vision test. Quality of life.*

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Resumen

Objetivo: *Desarrollar recomendaciones basadas en la evidencia para el diagnóstico, la rehabilitación funcional y el seguimiento de personas con baja visión entre 7 y 18 años y mayores de 18 años.* **Método:** *Se conformó el grupo desarrollador con la participación de pacientes. Se hizo una búsqueda sistemática yse evaluaron la calidad de los estudios y el cuerpo de la evidencia. Las recomendaciones se hicieron utilizando Grade Evidenceto Decision Framework, las cuales fueron validadas con expertos nacionales.* **Resultados:** *En el apartado de diagnóstico, la calidad de la evidencia respecto a agudeza visual y campo visual fue baja; en estas pruebas se generó recomendación débil a favor del uso de la cartilla Early Treatment Diabetic Retinopathy Study y de microperimetría, respectivamente. En cuanto a sensibilidad al contraste, la calidad de la evidencia fue muy baja y se da recomendación débil a favor del uso de la prueba de Pelli-Robson. En cuanto al tratamiento, la calidad de la evidencia respecto a ayudas ópticas electrónicas fue moderada y se da recomendación débil a favor para su uso. En cuanto a las lupas manuales, electrónicas, telescopios y tabletas, la calidad de la evidencia fue muy baja y se da recomendación débil a favor para su uso. En el apartado de rehabilitación, la calidad de la evidencia para personas mayores de 16 años fue moderada y se da recomendación fuerte a favor de la rehabilitación multidisciplinaria y multicomponente, mientras que la calidad de la evidencia respecto a rehabilitación para personas entre 7 y 16 años fue baja y se da recomendación fuerte a favor de su uso. La calidad de la evidencia para intervenciones de terapia ocupacional fue baja y se da recomendación débil a favor de su uso. Con relación a psicología, la calidad de la evidencia para las intervenciones de este tipo fue baja y se da recomendación fuerte a favor de su uso. Finalmente, la calidad de la evidencia para el uso de escalas de seguimiento fue alta y se da recomendación fuerte a favor del uso del National Eye Institute Visual Function Questionnaire-25.* **Conclusiones:** *Las recomendaciones emitidas permiten dar un lineamiento general para el manejo integral de la baja visión, reduciendo la variabilidad indeseada en el diagnóstico, la rehabilitación y el seguimiento.*

Palabras clave: *Baja visión. Guía de práctica clínica. Rehabilitación. Ayudas ópticas. Pruebas de visión. Calidad de vida relacionada con la salud.*

Introduction

Low vision is part of the functional spectrum of people with visual disability including moderate and severe visual deficiencies according to the best corrected visual acuity (VA) and visual field. It was reported that back in 2020 nearly 295 million people developed this condition, which amounts to 3.74% of the entire world population, with 37.4 cases reported for every 1000 people[1]. Within this vision loss category, 260 million people had a moderate visual disability (VA between 20/60 and 20/200) and 34.8 million had severe visual disability (VA between 20/200 and 20/400)[2]. That same year, in the Central American region, where Colombia is located, 34 million people were reported to have vision loss of whom 1.3 million were blind[3]. By 2050, taking into account population aging, the growth, and progressive urbanization of countries, among other determinant health factors, it has been anticipated that, at least, 474 million people will be living with this type of visual disability[3]. The main causes reported in 2021 were undercorrected refractive errors (86.1 million cases) and unoperated cataracts (78.8 million cases)[4].

Back in Colombia, preliminary reports of the 2018 population census report a national prevalence of this disability around 7.1% (3,065,361 Colombians have some disability of which 1,784,372 correspond to severe or total disability) and difficulties like not being able to see around them, near or far amount to 18.7%. Based on these numbers, visual disability could rank second in terms of frequency within the population with disabilities in the country[5].

Low vision can impact quality of life negatively and equitable access to education and employment, among others. This has considerable financial implications for people, families, and communities[1]. Visual disability usually compromises independent mobility and walking, decreases mental welfare, increases the probability of falls and car crashes, increases the need for social care and, ultimately, and leads to higher mortality rates[6].

In Colombia, there are not practical guidelines for the management of visual disability with recommendations regarding the diagnosis, treatment, rehabilitation, and follow-up of people with low vision. Therefore, a team was set up to develop one. This article describes evidence-based recommendations for the diagnosis of visual deficiencies in the low vision range, treatment with optical aids, functional vision rehabilitation, and follow-up of people with low vision between 7 and 18 years and in patients > 18 years, which provides support for the health-care personnel, people with low vision, their families, and the Colombian health-care system for the comprehensive management of people with this condition to improve the health-care process, its functionality, and participation.

Methods

These clinical practice guidelines were designed based on the Colombian Methodological Guidelines by the Ministry of Health[7].

A group was set up to develop the guidelines with an optometrist leader with a master's degree in public health and in policies and practices regarding development. Furthermore, a physiatrist methodological leader and a clinical epidemiologist experienced in evidence-based clinical practice guidelines. The remaining members of the clinical group were an optometrist with a master' degree in public health, an optometrist specialized in low vision, a health-care administrator with a master's degree in epidemiology, and a physician specialized in neuropsychological rehabilitation.

The methodological group including a physician with a PhD in health-care policies and a physiatrist with a master's degree in clinical epidemiology, both expert in systematic reviews, meta-analyses, evidence-based clinical practical guidelines drafting, and in the Grading of Recommendations, Assessment, Development, and Evaluations (GRADE methodology).

The group that developed the guidelines had permanent support from a specialist in archives and information sciences trained at the Iberoamerican Cochrane Center. External clinical experts from different colleges, hospitals, and organizations across the country were summoned. A group of 22 people with low vision and support from their relatives was built. Each of the participants signed a conflicts of interest statement form and they were approached according to the methodological manual.

This project had institutional backing from Research and Innovation Management from the Universidad Santo Tomás de Bucaramanga in full compliance with the 1993 Resolution 008430 enacted by the Colombian Ministry of Health on health-care research standards.

Objective of the guidelines

The aim of the study was to develop evidence-based recommendations systematically for the diagnosis, functional rehabilitation, and follow-up of people with low vision between 7 and 18 years of age and > 18 years, which can provide support for health-care personnel and patients for the comprehensive management of people with this condition to improve their participation and functionality, and optimize the health-care process within the Colombian Health System.

Population

The guidelines develop recommendations for the comprehensive care of patients between the ages of 7 and 18 years, and > 18 years with permanent moderate and severe visual impairment of the functional spectrum of visual disability who have visual function alterations that complicate or prevent performing activities of daily living, occupation, and participation.

These guidelines do not include patients with the following characteristics:

- < 7 years of age
- With mild visual impairment since both their visual condition and procedures regarding management are different to those performed in the context of low vision
- Surgical treatments are not included.

Drafting the questions while defining and rating the outcomes

According to the scope, purposes, and objectives of the guidelines, questions were asked with the Population, Exposure or Intervention, Comparison, Outcomes, and Time (PECOT) strategy. Afterward, the group developing the guidelines and the patients rated the outcomes independently following the GRADE system[8]. Each outcome was graded from 1 to 9. Medians between 7 and 9, 4 and 6, and 1 and 3 were categorized as critical, non-critical yet significant, and unimportant, respectively[8]. Other clinical practice guidelines on low vision or visual disability were looked for and assessed. Furthermore, additional information was considered using the following criteria: availability of the full text in English or Spanish, target population of patients > 7 years with low vision and whose overall score with the Appraisal of Guidelines for Research and Evaluation II (AGREE II) tool was > 40%[9]. The inclusion and exclusion criteria were used to select the clinical practice guidelines. They were assessed using the AGREE II too by two independent investigators[9]. A total of three clinical practice guidelines were included that met the selection criteria.

Systematic search for evidence

The systematic search for evidence included the following databases: Cochrane Library, Embase, PubMed, Bireme, and Epistemonikos. The members of the development group conducted manual and free searches through Current Controlled Trials, Trip Database, and Google Scholar from the beginning until

November and December 2019. Furthermore, they were updated between June and July 2020. MeSH terms were defined and selected, systematic reviews and meta-analyses were looked for and, additionally, clinical trials and observational studies.

Study selection and evidence description

The group that developed the guidelines was categorized into three groups including, at least, one or two thematic experts and a methodological one. Eligibility criteria were established for the articles regarding the methodological design, population, intervention, and minimum quality characteristics. Studies were excluded because they did not answer the PECOT question or because there was better or more recent evidence. The study selection by title and abstract regarding the treatment questions and scales was conducted by the three different epidemiologists who selected the studies that should be fully read. The thematic experts conducted the matrix of the characteristics of the studies with the variables number of study, title, author, journal, year, population, intervention or exposure, comparator, and outcomes. Afterward, all selected studies were described.

Quality of evidence

The quality of evidence of controlled clinical trials was supervised based on criteria defined by Cochrane: inadequate generation of the sequence, lack of allocation concealment and random allocation to therapy, masking, and loss to follow-up. Regarding systematic reviews and meta-analyses, the A Measurement Tool to Assess Systematic Reviews tool was used[10]. Regarding primary diagnostic studies, the Quality Assessment of Diagnostic Accuracy Studies tool[11] was used. Regarding primary observational studies, the Strengthening the Reporting of Observational studies in Epidemiology methodology was used[12]. Assessment of the quality of the body of evidence was conducted based on the concepts the GRADE methodology. The assessment of the quality of evidence is associated with confidence that the true effect is close to the estimated effect. Therefore, it was defined in four levels: very low, low, moderate, and high. The score was obtained after grading each outcome. The systematic reviews of clinical trials begin scoring with high quality (level 1) while reviews of observational studies begin with low quality (level 4). The aspects that can lower quality are bias risk, result inconsistency and inaccuracy, indirect evidence, and publication bias. One point (–1) or two points (–2) are subtracted based on severity and damage for each aspect. Although observational studies can lower quality due to the aspects mentioned, they can also increase quality when certain favorable methodological aspects are included. The three factors that can increase quality are the presence of a large effect size (relative risk, > 2.0 or < 0.5), the evidence of a dosage-response relation gradient, and the lack of residual bias or confounding factors[13-18]. Most quality of evidence for these guidelines was low or very low quality of evidence.

From evidence to recommendation

Following the GRADE system, recommendations were drafted based on these judgments: problem is a priority, desirable effects, unwanted effects, certainty of evidence, values, balance between sought and unsought effects, required resources, certainty of evidence of resources required, cost-effectiveness, equity, acceptability, and feasibility. With these criteria, a table was built including the summary of judgments. Furthermore, the direction and strength of the recommendation were decided[19].

The strength of recommendations was discussed with the entire group designing the guidelines, and was ranked as “strong for” or “strong against,” “weak for,” and “weak against.” Two socializations were made with experts: one at the beginning of the process to present the methodology and the questions of the guidelines, and the other at the end to present recommendations. The experts’ observations were analyzed and the suggestions discussed within the group. Eventually, the pertinent were accepted.

Results

The health-care process of people with low vision requires a comprehensive approach including different interventions and professionals are included who, all together, contribute to lessen the experience of the disability of people with low vision. The recommendations issued guide the health-care process of the health professionals regarding diagnosis, management, therapy, functional rehabilitation, and clinical follow-up of people with low vision > 7 years. Furthermore, they guide the patients and their caregivers regarding health-care aspects the recommendations, their quality of evidence, strength, and degree of recommendation as shown in table 1.

Diagnostic recommendations

Although it is widely accepted, moderate and severe visual impairment (low vision) in best corrected visual

Table 1. Synthesis of quality of evidence, strength, and direction of the recommendations

Intervention	Recommendation				Quality of evidence			
	Strong	Weak	For	Against	Very low	Low	Moderate	High
Visual acuity assessment is advised based on the ETDRS chart (Early Treatment Diabetic Retinopathy Study) to detect alterations and/or deficiencies in far visual acuity in people > 7 years old with visual impairment		■	■			■		
Assessment of contrast sensitivity is advised with the Pelli-Robson test to detect alterations and/or deficiencies in contrast sensitivity in people > 7 years old with visual impairment in addition to measuring visual acuity		■	■		■			
Use of microperimetry in assessing the visual field is advised to detect visual field alterations in people > 7 years old with visual impairment		■	■			■		
Use of electronic optical aids is advised for people < 16 years with low vision like closed-circuit TV mounted on stands, portable electronic devices with a mouse, and not mouse-based hand-held electronic devices (portable electronic magnifier) to improve reading speed, reading time, visual acuity reading, and quality of life		■	■				■	
Use of handheld electronic magnifiers, telescopes, and tablets is advised for children and teens with low vision between the ages of 7 and 16 years old to improve reading speed, visual acuity, reading comprehension, learning, and activities of daily living, participation, and quality of life		■	■		■			
Multidisciplinary and multicomponent vision rehabilitation is advised in low vision to improve health-and-vision-related quality of life, basic and instrumental everyday life activities, and reading capacity	■		■				■	
Rehabilitation interventions including physical training, group programs and training in the use of optical and non-optical devices is advised for people between 7 and 16 years with low vision to improve performance of activities of daily living, participation, and quality of life	■		■			■		
Inclusion of occupational therapy interventions in vision rehabilitation programs regarding low vision is advised to improve reading speed, performing basic and instrumental everyday life activities, and social participation		■				■		
There is no evidence for or against tele rehabilitation regarding low vision that would enable us to make a recommendation								
Group and/or individual psychology interventions based on self-management educational programs, problem-solving interventions, therapies based on peer support, or a combination of several strategies is advised in people > 16 years with low vision to improve vision-related quality of life, self-efficiency, self-esteem, depression symptoms, or depression. Also, to improve independence in activities of daily living	■		■			■		
Use of the National Eye Institute Visual Function Questionnaire scale (NEI VFQ) is advised for the early assessment and follow-up of vision-related quality of life in patients > 18 years with low vision	■		■					■
Total	4	6	10	1	2	4	2	1

acutiy (VA) can be accompanied by other deficiencies in color vision, contrast sensitivity, eye glare, and visual fields or depth vision. For the purpose of these guidelines, we followed the proposals established by the International Classification of Diseases, version 10 reviewed, and version 11 of the World Health Organization that define low vision based on the classification of VA ranges. Therefore, low vision is understood as a best corrected VA worse than 20/60, but equal to or better than 20/400 in the best eye or loss of visual field corresponding to < 20° in the best eye with the best possible correction[20]. In addition, visual disability for near vision is defined when VA is < N6 or N8 at 40 cm (type and size expressed in typographic fonts and *Times Roman*), and when VA for far sight is worse than or equal to 20/40 with the best correction[21].

Visual acuity

Assessment of VA for far sight is advised with the Early Treatment Diabetic Retinopathy Study (ETDRS chart) compared to the assessment with the Snellen chart to detect alterations or deficiencies in VA for far sight in people > 7 years with low vision. (The quality of evidence regarding VA was low. Weak recommendation is given favorable to the use of the ETDRS chart).

The systematic review conducted by Chou et al.[22] back in 2009 identified six studies[23-27], but none of them directly compared the diagnostic accuracy of VA assessment to the ETDRS chart and Snellen regarding sensitivity, specificity, likelihood ratio or predictive values. In the systematic review of the medical literature available on VA screening-of moderate quality – «all» methods are advised[22]. In the five articles of agreement/reproducibility, the ETDRS test performed better thanks to having less variability compared to the Snellen chart, although the use of the ETDRS test[23-27] involves longer application times. Despite of this and considering the lower variability, greater agreement, and reproducibility of the ETDRS compared to the Snellen test, and the design characteristics and its logarithmic progression, it is advised to add this test to determine and quantify the capacity of VA discrimination more accurately, and therefore be able to identify whether the person is in the low vision range or not. Probably when balancing sought and unsought effects, the implementation of the ETDRS test is promoted because the only unwanted effect reported had to do with the additional time associated with its use.

Implementation considerations

The implementation of the ETDRS charts and the decision to update clinical care involves costs in materials and personnel training[27]. The ETDRS chart costs between USD 40 and USD 45. The backlit ones costs between USD 2500 and USD 4000[28]. Similarly, the health-care professional training process should be reinforce for the proper use and interpretation of ETDRS.

Contrast sensitivity

Assessment of contrast sensitivity is advised with the Pelli-Robson test to detect contrast sensitivity alterations or deficiencies in people > 7 years with visual impairment as an additional way to determine VA. (The quality of evidence regarding contrast sensitivity was very low. Recommendation favorable to the use of the Pelli Robson test is weak.) Mars test is advised for the lack of the Pelli-Robson test.

Five articles on contrast sensitivity assessment with the Pelli-Robson test were identified, the Mars test or the Functional Acuity Contrast Test in people with eye conditions that can cause visual deficiencies in moderate-to-severe categories[29-33]. No systematic reviews or controlled clinical trials were found. None of the studies made direct comparisons of diagnostic precision or the discrimination capacity of the contrast sensitivity tests among them. Both the Pelli-Robson and Mars tests had similar reproducibility and excellent agreement. However, the former had an area under the Receiver Operating Characteristic curve of 0.89, which is indicative of an excellent discrimination capability[32]. The effects of implementing the use of the Pelli-Robson test basically translate into an improved visual and ocular healthcare since it allows us to determine and quantify visual quality with greater precision in people with visual deficiencies in the low vision range, which is essential to perform activities of daily living. Its clinical significant was revealed in studies that confirm that many ophthalmological conditions and disorders such as glaucoma[34], macular degeneration[35], diabetic retinopathy[36], and cataracts[37], among others, show measurable contrast sensitivity losses even in cases where VA can be in the normal range[38].

Implementation considerations

Implementation of the Pelli-Robson test and the decision to update clinical care involve cost in equipment and personnel training. The cost of acquiring the Pelli-Robson and the Mars tests go from USD 1000 to USD 1500, and from USD 200 to USD 300, respectively.

Visual field

Visual field assessment is advised through microperimetry compared to digital campimetry examination to detect visual field alterations in people > 7 years with low vision. If this test is not available, the use of digital campimetry is advised. (The quality of evidence regarding visual field was low. Recommendation favorable to the use of microperimetry was weak).

Seven diagnostic precision studies were selected[39-45], six of which are favorable to microperimetry and one does not identify any differences between microperimetry and digital campimetry. The reproducibility reported is better for microperimetry compared to digital campimetry[39,41,45]. It was found that the area under the curve for microperimetry was greater compared to digital campimetry (0.0190 and 0.735, respectively)[39,42]. The MP-3 microperimeter has a better structure-function relationship and correlates better with optical coherence tomography[40]. The use of microperimetry allows us to determine the patient's visual field alterations and improve the prognosis of the visual condition since the earlier visual field deficiencies are found, the better they can be treated and rehabilitated. Regarding the presence of ocular diseases, several studies have confirmed their utility to assess visual function in patients with eccentric or unstable fixation[44] and analyze different retinla diseases such as age-related macular degeneration and glaucoma[46].

Implementation considerations

Although microperimetry is used in different settings not only for diagnosis, but also for vision rehabilitation, the process of training human resources, as well as its use and interpretation should be reinforced since it is a test that is just beginning to be used in Latin America and Colombia.

Treatment recommendations

Electronic optical aids

Use of electronic optical aids is advised for people > 16 years with low vision like closed-circuit TV mounted on stands, portable electronic devices with mouse and mouse-free hand-held electronic devices (like the portable electronic magnifier) to improve reading speed, time, and acuity, and quality of life. (The quality of evidence regarding electronic optical aids is moderate. Recommendation given favorable to its use was weak).

There is a wide variety of priority back-up products or low vision aids considered like any other device that allows somebody with low vision to improve his visual performance. The evidence found supported mainly by the meta-analyses conducted by Virgili et al.[47] shows that reading speed can improve with closed-circuit TV mounted on a stand compared to hand-held or stand-mounted magnifiers (evidence of low quality). On the other hand, reading time is longer with electronic compared to ordinary optical aids devices, and they are also easier to use (moderate evidence). In addition, reading time is longer with electronic devices with a camera mounted on a mouse compared to traditional optical aids (moderate evidence). Finally, only one study reported showed that reading time is longer at 1 month when the patients were prescribed closed-circuit TV at the beginning and not only optical aids. Regarding good clinical practice, it is advised that if electronic aids are unavailable, optical aids such as microscopes, magnifying glasses (any available), and transparent filters should be used to improve reading speed, time, and acuity[48-50].

Implementation considerations

Electronic aids are not readily available in the market in Colombia. Most of them need to be imported and are not covered by the health-care system[51]. Price goes from USD 190 for electronic magnifiers, USD 280 for closed-circuit TVs, and USD 2796 for head-mounted devices. Studies published in other countries show that they are cost-efective[51]. Patients would accept interventions with electronic optical aids if they could assess their effectiveness, and accept them based on size, portability, and accessibility.

Manual or electronic magnifiers, telescopes, and tablets

Use of manual or electronic magnifiers, telescopes, and tablets is advised in children and teenagers with low vision between the ages of 7 and 16 years to improve reading speed, VA, reading comprehension, learning, perform activities of daily living, participation, and quality of life. (The quality of evidence regarding hand-held or electronic magnifiers, telescopes, and tablets was very low. Recommendation given favorable to its use was weak). The optical aids commonly used in people between 7 and 16 years include hand-held magnifiers with a stand (with and without lighting), magnifying reading glasses with high dioptric power of up

to + 20.00 diopters, bifocals > + 4.00 diopters, long-range monocular or binocular telescopes, pre-mounted lens system to magnify at a greater distance, and electronic magnification using closed-circuit TV[52]. Other devices that have become popular in educational settings are screen magnifier software and computer-operated screen reading on desktop and laptop computers or tablets, and devices including displays to show enhanced text[53]. Mobile devices like tablets have become widely available as conventional tools and are also used in some schools[47].

For the sake of good practice, optical aids such as optical magnifiers, stand magnifiers without lighting and electronic devices for near vision, telescopes for far vision, and tablets with their corresponding software and applications should be used, as well as non-optical aids like filters, reading stands, large print, and lamps to improve reading and comprehension, learning, independence in activities of daily living, and social participation[54-57].

Implementation considerations

The Colombian Health-care System should be reinforced since back-up devices are not included in the Health-care Benefit Plan. In addition, there are not enough human resources available trained in the process of prescription and training in this category of aids, and the ones that do exist are in large cities[58]. Barriers are recognized regarding cost and availability of aids in the country, as well as fear of stereotypes associated with disability that can cause these people to oppose the use of optical aids.

Rehabilitation recommendations

Rehabilitation in people > 16 years

Multidisciplinary and multicomponent rehabilitation is recommended in people with low vision to improve their health-and-vision-related quality of life, basic and instrumental activities of daily living, and their reading ability. (The quality of evidence regarding rehabilitation of people > 16 years was moderate. Recommendation given favorable to multidisciplinary and multicomponent rehabilitation was strong).

Multicomponent rehabilitation consists of vision rehabilitation programs including education and information exchange on low vision, problem-solving skill training, relaxation techniques, and teaching/training on the use of optical and non-optical low vision devices[59]. These rehabilitation services include a wide variety of services, additional elements beyond the mere prescription, and training in the use of optical devices such as training in orientation and mobility, occupational therapy, the acquisition of skills for independent living, expert advice and prescription, group activities, training regarding aid devices, and educational and employment advice to facilitate continuous participation in various activities and promote well-being[60]. This multicomponent model is provided through integrated or multidisciplinary services[61-63]. We should mention that there is high variability in the type of interventions, comparators, and outcomes. The recommendations issued are based on four systematic reviews[63-66] and four individual studies[67-70].

Implementation considerations

The dynamics under which rehabilitation services are rendered in Colombia should be reinforced since the fragmentation of the Social Security System does not allow comprehensiveness, continuity, or opportunity of processes[71]. Similarly, there are not enough health-care institutions equipped to implement these recommendations associated with multidisciplinary and multicomponent vision rehabilitation processes, and the ones that do exist are located in large cities only. In addition, there are not enough skilled human resources for this process[58].

Rehabilitation of people between 7 and 16 years

Rehabilitation is advised in people between 7 and 16 years with low vision. These interventions should include physical training and group programs and training in the use of optical devices to improve performance in activities of daily living, participation, and quality of life. (The quality of evidence regarding rehabilitation for people between 7 and 16 years was low. Strong recommendation is given favorable to its use).

Vision rehabilitation services adjust to a wide variety of different models; some approach only the person's functional needs with an emphasis on provision of optical and non-optical aids while others take a more holistic approach. The recommendation is based on the systematic review conducted by Elsman et al.[55] in which the studies included show that providing physical training or sports camps is probably effective to improve physical performance. Training in path-following tasks was effective to improve fine motor skills,

regardless of the use of visual aids. Group programs to improve psychological results showed mixed results[55]. Mohamed et al.[72] found larger effect sizes, in which an educational group program turned out to be better regarding activities of daily living, anxiety, depression, and self-esteem. However, this study did not include a comparison group. Sports camps seem to effectively improve psychological results. Rehabilitation conducted in low vision rehabilitation centers was effective, especially combined with a home-based program with community support. Provision of low vision devices was effective to improve function and development, although these studies did not use a control group. Although limited in number, it is likely that interventions to improve mobility can also be effective.

Implementation considerations

The health-care system should be reinforced because fragmentation in health-care does not allow comprehensiveness, continuity, and opportunity for vision rehabilitation processes. In addition, the number of health-care centers and health professionals trained to implement these recommendations associated with multidisciplinary rehabilitation processes should go up[51].

Occupational therapy

Inclusion of occupational therapy interventions in vision rehabilitation programs targeting people with low vision is recommended to improve reading speed, performance of basic and instrumental everyday life activities, and social participation. (The quality of evidence regarding occupational therapy interventions was low. Recommendation given favorable to its use was weak).

Based on the individual needs and characteristics of the person with low vision, occupational therapy interventions or sessions in adults should be initially provided individually with short but more frequent sessions including adaptation to low vision devices[72]. In certain cases, based on the need for socialization and interaction with peers, information exchange and skill acquisition to solve problems for people with low vision, occupational therapy interventions can be performed in groups also with short sessions but more often including adaptation to low vision devices[59]. In the case of group occupational therapy, rehabilitation sessions should be performed in small groups and for several weeks[59]. Occupational therapy professionals working with elderly people with low vision should expand their approach beyond basic and instrumental everyday activities. Leisure and social participation are important occupational areas essential for health and well-being. Therefore, more emphasis should be put on these aspects that are often overlooked. In patients whose objective, it is to improve reading performance, occupational therapy professionals should include training with electronic magnifiers and adequate lighting needs. Furthermore, instructions to learn to use optical magnifiers and eccentric vision should be included, although the evidence of these intervention strategies is limited[73].

Implementation considerations

Occupational therapy should be reinforced regarding specific training in low vision. The role and significance of occupational therapy should be emphasized so it becomes a common part of the health-care teams responsible for performing vision rehabilitation programs.

Telerehabilitation in low vision

There is no evidence in favor or against the use of telerehabilitation in vision rehabilitation processes in people with low vision so as to make any recommendations[74].

Psychology

Psychological group or individual interventions are advised in people > 16 years with low vision based on problem-solving self-management, educational, and intervention programs, peer-support-based therapies, or a combination of several strategies to improve vision-related quality of life, self-effectiveness and self-esteem, reduce depressive symptoms, and improve independence in everyday life activities[64,76]. (The quality of evidence regarding psychological interventions was low. Recommendation given favorable to its use was strong). Psychological therapy or methods to improve vision, compared with active control groups, report little benefit on vision-related quality of life. The type of rehabilitation varied among studies, and, also, within the intervention groups. However, benefits were detected even compared to active control groups. Most studies included on low vision rehabilitation had a brief follow-up[76]. At present, there is only limited evidence on the effectiveness of psychosocial interventions in the field of low vision. More well-designed trials are needed paying special attention to interventions adapted to the needs of elderly patients with low vision.

Implementation considerations

Comprehensive care programs should be created to better coordinate psychological interventions with the interdisciplinary team. Similarly, an essential factor regarding implementation is the patients or their caregivers' resistance to accept intervention in psychology.

Follow-up recommendations

Follow-up scales

Use of the NEI-VFQ (National Eye Institute Visual Function Questionnaire) is advised for the early assessment and follow-up of vision-related quality of life in patients > 18 years with low vision. (The quality of evidence regarding the use of follow-up scales was high. Recommendation given favorable to the use of NEI-VFQ was strong).

The studies that evaluated the psychometric properties of NEI-VFQ through the classic theory of the test reported adequate psychometric properties and good internal consistency. The test-retest stability was adequate in all validations while the validity of the construct and the content was appropriate in several studies[77-86]. There are two different versions: the original one including 51 items and the short one including 25. NEI-VFQ-25 is the scale with the largest number of validations reported and is considered the standard of reference[87]. Its validity and reliability remain regarding the original version[88]. This scale measures the influence of visual disability and visual symptoms in the domains of physical and mental health of the population in addition to other domains oriented to tasks associated with daily visual performance[89].

Psychometric properties show that it is a valid and reliable scale, constant in time, and sensitive to change. Using the scale translated, adapted, and validated to the Colombian context is advised. The driving subscale answers should be interpreted with caution since several patients with low vision do not drive according to the legislation, which can change from one country to the next.

Implementation considerations

The scale should be known and used by health-care professionals, especially those involved with visual care. Furthermore, they should train in scoring and result interpretation. It can take additional time in the office to be used.

Conclusions

These clinical practice guidelines have produced recommendations based on the best evidence available regarding the diagnosis, rehabilitation therapy, vision rehabilitation, and clinical follow-up of people > 7 years with low vision. The GRADE approach allowed us to develop a complete, explicit, and transparent methodology to make several specific and useful recommendations on how to approach people with low vision in different health-care stages.

The drafting of the guidelines allows us to make an impact by reducing unwanted variability in the diagnosis, rehabilitation, and follow-up of people with low vision. The guidelines provide general outlines for the comprehensive management of this condition and use of therapies and clinical approaches to optimal quality standards, which is expected to translate into a less potentially negative impact of low vision as a type of visual disability in the patients' activities of daily living.

The recommendations issued are essential to cope with the current challenges of mankind since with an aging world population, visual disability has become a public health problem regarding measures to prevent and detect, in a timely manner, visual deficiencies capable of causing permanent low vision, access to vision rehabilitation services and programs, supply and demand of aid devices, and better knowledge of the functional diversity and quality of life of people with low vision. Therefore, a key aspect to promote the rights of people with visual disability is to improve their functional abilities by increasing access to vision rehabilitation services and creating more inclusive environments by reinforcing inclusive policies and laws. Therefore, like Lancet Global Health Commission on Global Eye Health puts it, we need to make sure that vision rehabilitation should be considered a comprehensive component of eye care services in our quest for universal health-care coverage[1]. Still, people who seek vision rehabilitation face a variety of challenges including, among others, scarce services, physical barriers, inadequate skills and knowledge by the health-care team, and lack of knowledge by other sectors of society. These barriers contribute to the overall estimate that only 15% of people could probably benefit from these services if they had access to them[2].

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Conflict of interests

None.

Ethical disclosures

Protection of humans and animals. The authors declare that no experiments on humans or animals have been performed for this research.

Confidentiality of data. The authors declare that no patient data appear in this article.

Right to privacy and informed consent. Right to privacy and informed consent. The authors have obtained the approval of the Ethics Committee for the analysis and publication of routinely obtained clinical data. The informed consent of the patients was not required because this was a retrospective observational study.

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